* 8M/10M

UNIT-2

MINIMIZATION OF DFA USING TABLE FILLING METHOD

Equivalance of two states

* Indistinguishable state

Two states P and Q of DFA are equivalent (Indistinguishable) if and only if $\delta(p, w)$ and $\delta(Q, w)$ are final states or $\delta(P, w)$ and $\delta(Q, w)$ are non final states

$\delta(p, w) \in F$ and $\delta(q, w) \in F$ or

$\delta(p, w) \notin F$ and $\delta(q, w) \notin F$

* Distinguishable state

Consider $\delta(P, w)$ and $\delta(Q, w)$ among these two (if one is final state) that is- $\delta(P, w) \in F$ and $\delta(P, w) \notin F$ (another nonfinal) the P and q are not equivalend and are called Distinguishable states.

Table filling algorithm

Used to find Indistinguishable & distinguishable

step 1: Identify the initial marking

for each pair $(P, Q)$ where $P \in Q$ and $q \in Q$

if $P \in F$ and $q \notin F$ or vice versa then pair $(p, q)$ is called distinguishable pair

step 2: Identify the subsequent marking

for each pair $(P, Q)$ and for each $a \in \Sigma$

$\delta(p, a) = r$ and $\delta(q, a) = s$

If the pair $(r, s)$ is marked as distinguishable then the pair $(P, Q)$ is also distinguishable & Delete it as X

step 3 Repeate step 2 until no previously unmarked pairs are marked

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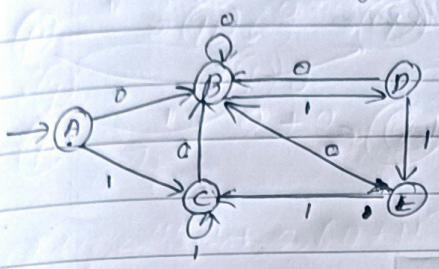
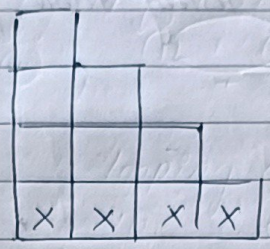
Minimize the following DFA using table filling method.

(1)




remove 1st state

consider distinguish. (one F one N.F)

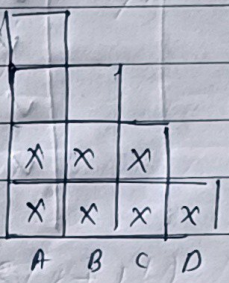
consider not marked states

removed last state to avoid redundancy

| | 0 | 1 | |
|---|---|---|---|
| (B,A) | (B,B) | (D,C) | ✗ not marked |
| (C,A) | (B,B) | (C,C) | pair does not exist } in box |
| (C,B) | (B,B) | (C,D) | pair does not exist } |
| (D,A) | (B,B) | (E,C) | ✓ |
| (D,B) | (B,B) | (E,D) | ✓ marked |
| (D,C) | (B,B) | (E,C) | ✓ |

↳ (D,A), (C,A) (D,1), (C,1)

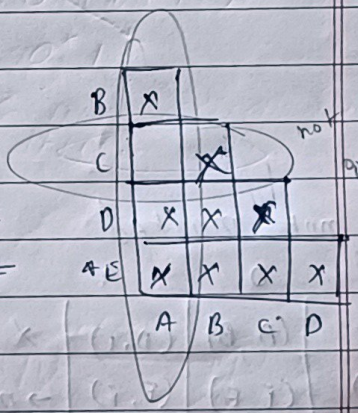
so mark (D,A) (D,B) (D,C)



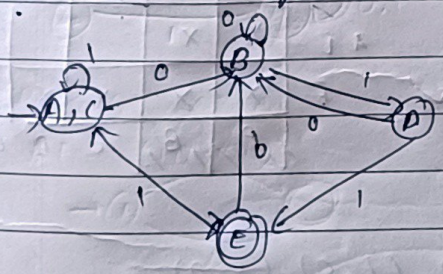

consider not marked

| | 0 | 1 | |
|---|---|---|---|
| (B,A) | (B,B) | (D,C) | ✓ |
| (B,A) | (B,B) | (C,C) | pair doesn't not exist |
| (C,B) | (B,B) | (C,D) | ~~pair does not exist~~ |

↳ (x,D) same as (D,C) as marked consider




Now consider complete marked states B, D, E



Minimize the DFA using table filling algorithm

| | 0 | 1 |
|---|---|---|
| → A | B | C |
| B | A | D |
| * C | E | F |
| A D | E | F |
| * E | E | F |
| * F | F | F |

| | A | B | C | D | E |
|---|---|---|---|---|---|
| B | | | | | |
| * C | X | X | | | |
| * D | X | X | | | |
| * E | X | X | | | |
| F | | | X | X | X |

| | 0 | 1 | |
|---|---|---|---|
| (B, A) | | | ✓ |
| (C, A) | | | ✓ |
| (E, B) | | | ✓ |
| (D, A) | | | ✓ |
| (D, B) | | | ✓ |
| (D, C) | | | |

| | 0 * | 1 * | * |
|---|---|---|---|
| (E, A) | | | ✓ |
| (E, B) | | | ✓ |
| (E, C) | | | |
| (E, D) | | | |
| (F, A) | | | |
| (F, B) | | | |
| (F, C) | | | ✓ |
| (F, D) | | | ✓ |
| (F, E) | | | ✓ |

Now not marked

| | 0 | 1 | |
|---|---|---|---|
| (B, A) | (A, B) | (D, C) | X |
| (D, C) | (E, E) | (F, F) | → pair does not exist |
| (E, C) | (E, E) | (F, F) | → pair does not exist |
| (E, D) | (E, E) | (F, F) | → pair does not exist |
| (F, A) | (F, B) | (F, E) | ~~→ pair does not exist~~ ✓ |
| (F, B) | (F, A) | (F, D) | ✓ |

| | A | B | C | D | E |
|---|---|---|---|---|---|
| B | | | | | |
| * C | X | X | | | |
| * D | X | X | | | |
| * E | X | X | | | |
| F | X | X | X | X | X |

| | 0 | 1 |
|---|---|---|
| (B, A) | (A, B) | (D, C) |
| (D, C) | (E, E) | (F, F) |
| (E, C) | (E, E) | (F, F) |
| (E, D) | (E, E) | (F, F) |

(B, A), (D, C), (E, C), (E, D)

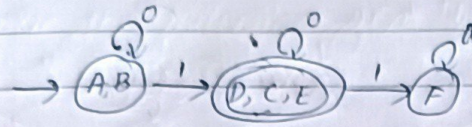
as F completely marked F is distinguishable state

as A is s.s (B, A) is s.s

~~and as C D E or F.s~~ un marked

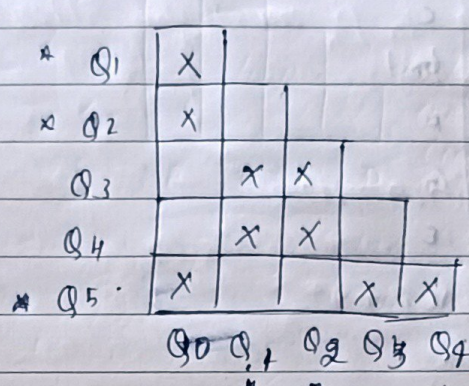
(D, C) (E, C) (E, D) → F.s

(D, C, E) indistinguishable.



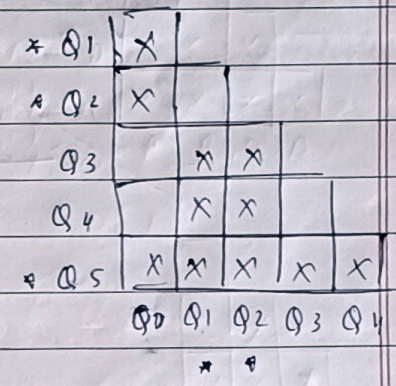

3.

| | 0 | 1 |
|---|---|---|
| → $Q_0$ | $Q_1$ | $Q_2$ |
| * $Q_1$ | $Q_3$ | $Q_4$ |
| * $Q_2$ | $Q_4$ | $Q_3$ |
| $Q_3$ | $Q_5$ | $Q_5$ |
| $Q_4$ | $Q_5$ | $Q_5$ |
| * $Q_5$ | $Q_5$ | $Q_5$ |

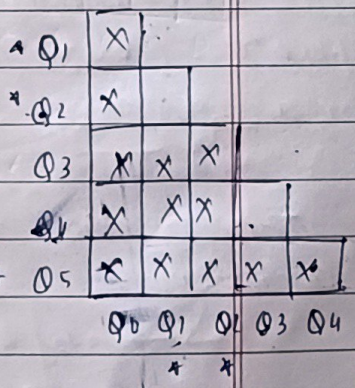



Now not marked

| | 0 | 1 | |
|---|---|---|---|
| $(Q_2, Q_1)$ | $(Q_4, Q_3)$ | $(Q_3, Q_4)$ | X |
| $(Q_3, Q_0)$ | $(Q_5, Q_1)$ | $(Q_5, Q_2)$ | X |
| $(Q_4, Q_0)$ | $(Q_5, Q_1)$ | $(Q_5, Q_2)$ | X |
| $(Q_4, Q_3)$ | $(Q_5, Q_5)$ | $(Q_5, Q_5)$ | pair does not exist |
| $(Q_5, Q_1)$ | $(Q_5, Q_3)$ | $(Q_5, Q_4)$ | ✓ |
| $(Q_5, Q_2)$ | $(Q_5, Q_4)$ | $(Q_5, Q_3)$ | ✓ |




| | 0 | 1 | |
|---|---|---|---|
| $(Q_2, Q_1)$ | $(Q_4, Q_3)$ | $(Q_3, Q_4)$ | X |
| $(Q_3, Q_0)$ | $(Q_5, Q_1)$ | $(Q_5, Q_2)$ | ✓ |
| $(Q_4, Q_2)$ | $(Q_5, Q_1)$ | $(Q_5, Q_2)$ | ✓ |
| $(Q_4, Q_3)$ | $(Q_5, Q_5)$ | $(Q_5, Q_5)$ | pair does not exist |

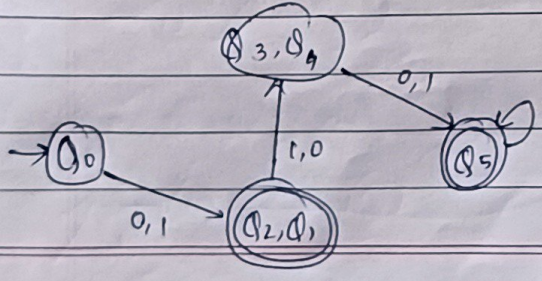



∴ $S = (Q_2, Q_1), (Q_4, Q_3)$

$D \cdot S = Q_5, Q_0$

here, consider unmarked pairs as indistinguishable

∴ $Q_5$, $Q_0$ are distinguishable.

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| δ | a | b |
|---|---|---|
| A | B | F |
| B | G | C |
| C | A | C |
| D | C | G |
| E | H | F |
| F | C | G |
| G | G | E |
| H | G | C |

Context free grammers and languages.

Monisha ate slowly
noun verb adverb

Grammer has 4 types

$G = (V, T, P, S)$ $V \rightarrow$ Variables or non terminals

$T \rightarrow$ terminals

$P \rightarrow$ Production or Rules.

$S \rightarrow$ state symbols

$\alpha \longrightarrow \beta$

$\alpha \in (V \cup T)^+$

$\beta \in (V \cup T)^*$

| Terminals | Non terminals |
|---|---|
| Keyword if, for, while | lowercase names - expression, statements |
| digits 0-9 | |
| lowercase letter a-z | Capital letters A-Z |
| Boldfaced words - | |

$\alpha \longrightarrow \beta$ $\begin{cases} \alpha \in (V \cup T)^+ \\ \beta \in (V \cup T)^* \end{cases}$

Chomsky Hierarchy

Types of grammer

Type 0 grammer

Type 1 grammer

Type 2 grammer → Context free grammer

Type 3 grammer

Type 0 grammer.

~~Phase~~ Phrase structured Grammer or unrestricted Grammer

$G = (V, T, P, S)$

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Type 1 Grammer

Context sensitive grammer or ε-free grammer

$\alpha \rightarrow \beta$ → can't have ε

length of $\beta$ must be atleast same as length of $\alpha$

$|\beta| \geq |\alpha|$

Type 2 Grammer

Context free Gramer

$\alpha \in (V \cup T)^*$

$A \rightarrow \alpha$

only 1 var → A; can have ε → α

$G = (V, T, P, S)$

Ex -

$S \rightarrow aB / bA / \varepsilon$

$A \rightarrow aA / b$

$B \rightarrow aB / a / \varepsilon$

Type 3 Grammer

Regular grammer

Right linear $A \rightarrow wB$ or $A \rightarrow w$

Left linear $A \rightarrow Bw$ or $A \rightarrow w$

Ex - R.L

$S \rightarrow aaB / bbA / \varepsilon$

$A \rightarrow aA / b$

$B \rightarrow bB / a / \varepsilon$

Ex - L.L

$S \rightarrow Baa / Abb / \varepsilon$

$A \rightarrow Aab$

$B \rightarrow Bb / a / \varepsilon$

Grammer $G = (V, T, P, S)$ is said to be Type 2 grammer or context free grammer are of the ~~uluma~~ form of $A \rightarrow \alpha$ where $\alpha \in (V \cup T)^*$ and A is a non-terminal or a variable

The symbol ε can be appear on the RHS of any production.

$S \rightarrow aB / bA / \varepsilon$

$A \rightarrow aA / b$

$B \rightarrow aB / a / \varepsilon$

~~Grammar from FA~~

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Grammer from FA

$M = (Q, \Sigma, \delta, q_0, F)$

$Q$

$Q = (q_0\ q_1\ q_2 \ldots)$

$\Sigma = (a_1\ a_2 \ldots)$

$\delta : Q \times \Sigma$ to $Q$

$q_0 \rightarrow$ start state

$G = (V, T, P, S)$

$V = (q_1\ q_2\ q_3 \ldots)$

$T = (a_1\ a_2\ a_3 \ldots)$

$q_0 \rightarrow$ S.S of C.F Gr

$\delta(q_p, a) = q_j$

$q_p \rightarrow aq_j$

Ex

$\delta(p, a) = q$

$p \rightarrow aq$

Obtain a grammer to generate string consisting of any no. of A's

$L = \{\varepsilon, aa, aaa \ldots\}$

$V = \{S\}$

$T = \{a\}$

S. symbol → S is id on this

transition

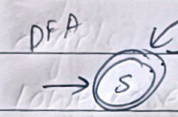
$\delta(S, a) = S$ → rules

CFG

$S \rightarrow aS$

$S \rightarrow \varepsilon$ S is the final state

DFA




$\delta(q_i, a) = q_j$

$q_i \rightarrow aq_j$

$S \rightarrow aS / \varepsilon$

mean S can be replaces by aS or $\varepsilon$

$S \Rightarrow aS$

$\Rightarrow a\ aS$

$\Rightarrow aa\ aS$

$\Rightarrow aaa\ \varepsilon$

as 3 a's we got now S is replaced by $\varepsilon$ that is $aaa\varepsilon$

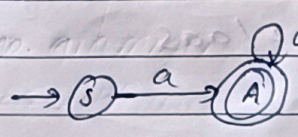
$aaa$

120 – 120 × 8.

Obtain the grammer to generate string consisting of atleast one a

$L = \{a, aa, aaa, \ldots\}$

DFA




$\delta(S, a) = A$

$\delta(A, a) = A$

A is the final

any no. of a's = $S \rightarrow \varepsilon / aS$

atleast 1 a = $S \rightarrow a / aS$

CFG

$S \rightarrow aA$

$A \rightarrow aA$ (or) $S \rightarrow a / aS$

$A \rightarrow \varepsilon$ F.s always have $\varepsilon$

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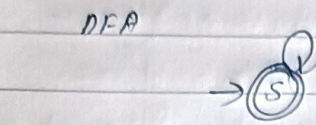
Obtain a grammer to generate strings consisting of any no. of a's and b's

$L = \{\varepsilon, a, b, ab, aab, abb, ---\}$

DFA




| |
|---|
| $\delta(S,a) = S$ |
| $\delta(S,b) = S$ |
| S is final state |

CFG

$S \to aS$

$S \to bS$ or $S \to \varepsilon / aS / bS$

$S \to \varepsilon$

① CFG

$S \to \varepsilon / aS$ any no. of a's

$S \to \varepsilon / bS$ any no. of b's

$S \to \varepsilon / aS / bS$

$S \to \varepsilon / aS / bS$

Obtain a grammer to generate string consisting of atleast 2 a's

$L = \{aa, aaa, aaaa, ---\}$

DFA

① CFG

$S \to aa / aS$

rules

here $V = \{S\}$

$T = \{a\}$

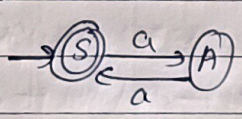
start symbol $\to S$

~~Obtain a~~

Obtain a grammer to generate string consisting of even no. of a's

$L = \{\varepsilon, aa, aaaa ---\}$

DFA




| |
|---|
| $\delta(S,a) = A$ |
| $\delta(A,a) = S$ |
| final state is S |

① CFG

$S \to \varepsilon / aaS$

CFG

$S \to aA$

$A \to aS$

$S \to \varepsilon$

~~or $S \to \varepsilon / aaS$~~

$\Rightarrow S \to aA / \varepsilon$

$A \to aS$

or $S \to \varepsilon / aaS$

Obtain a grammer to generate string consisting of any no. of a's and b's with atleast one a

$L = \{a, aab, abb ---\}$

① CFG rule

$S \to a / aS / bS$

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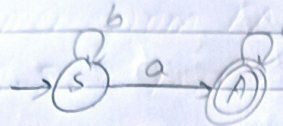
DFA




string consisting of any no of as and bs with atleast one a or atleast one b.

$S \to a/b/as/bs$

~~① CFG~~

~~$S \to a/b/as$~~.

## GRAMMER FROM REGULAR EXPRESSION

1) Obtain a grammer to generate strings of as and bs having a substring ab.

$(a+b)^* ab (a+b)^*$

$S \to A\ ab\ A$ here A means any no of as & bs

$A \to aA/bA/\varepsilon$ in eqn

2) Strings of as and bs ending with ab

$(a+b)^* ab$

$S \to A\ ab$ $\quad V = \{S, A\}$

$A \to \varepsilon/aA/bA$ $\quad T = \{a, b\}$

3) strings of as and b's starting with ab

$ab\ (a+b)^*$

$S \to ab\ A$

$A \to \varepsilon/aA/bA$

## GRAMMER FOR OTHER LANGUAGES

1) Obtain a grammer to generate the following language

$L = \{a^n b^n : n \geq 0\}$

$S \to \varepsilon/asb$

$asbs$

not $asbs$

bcz b not followed by any no of as $asbs$

$asbasbs$

a s b

$a(asb)b$

$aasbsbs$

$aa\varepsilon b\varepsilon$

2) Obtain the grammer to generate $L = \{a^n b^n \mid n \geq 1\}$

$S \to ab / aSb$ $\quad V = \{S\}$

$T = \{a, b\}$

$P = S \to ab / aSb$

S is the start symbol

3) $L = \{a^{n+1} b^n : n \geq 0\}$

| n | $a^{n+1} b^n$ |
|---|---|
| 0 | a |
| 1 | aab |
| 2 | aaabb |

$S \to a / aSb$

$V = \{S\}$

$T = \{a, b\}$

$P = S \to a / aSb$

S is the start symbol

$a\underline{S}b$

$a\,a\underline{S}b\,b$

$a\,a\,a\underline{S}b\,b\,b$

4. $L = \{a^n b a^n \mid n \geq 0\}$

$L = \{b, aba, aabaa \cdots\}$

$S \to b / aSa$

$S \Rightarrow a\underline{S}a$

$aa\underline{S}aa$

$aabaa$

5. $L = \{a^n b^{n+2}, n \geq 0\}$

$L = \{bb, abbb, abbbbb \cdots\}$

| n | $a^n b^{n+2}$ |
|---|---|
| 0 | bb |
| 1 | abbb |
| 2 | aabbbbb |

$S \to bb / aSb$

6. $L = \{a^n b^{2n} \quad n > 0\}$

$L = \{\varepsilon, abb, aabbbb \cdots\}$

$S \to \varepsilon / aSbb$

7.

2. What is the language generated by the following grammer.

$S \to 0A / \varepsilon$

$A \to 1S$

01S

0101S

$S \Rightarrow \varepsilon$

$S \Rightarrow 0A$

$\Rightarrow 01S$

$\Rightarrow 010A$

$\Rightarrow 0101S$

$\Rightarrow 01010A$

$\Rightarrow 0101015 \to 01010 1\varepsilon$

$L = \{(01)^n, \; n \geq 0\}$

(not $0^n 1^n$)

$L = \{a^{n+2} b^m \mid n \geq 0$ and $m > n\}$

| n | m | $a^{n+2} b^m$ |
|---|---|---|
| 0 | 1... | aabb* |
| 1 | 2... | aaabbb* |
| 2 | 3... | aaaabbbb* |

here for $n = 0$ $m > n$ given

$\therefore$ m can be 1, 2, 3 ...

$\therefore b^*$ taken

means zero also include

closure

$a(n \text{ no of } a \; n \text{ no of } b) b^*$

$S \to aAb$

$A \to aAb / ab$

$B \to bB / \varepsilon$

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$L = \{a^n b^m \mid n \geq 0, m \geq n\}$

| n | m | $a^n b^m$ |
|---|---|---|
| 0 | 1 | $bb^*$ |
| 1 | 2 | $abbb^*$ |
| 2 | 3 | $aabbbb^*$ |

equal no. of as equal no. of bs $b\,b^*$

$L = \{bb^*, abbb^*, aabbbb^* \ldots\}$

$S \rightarrow AB$

$A \rightarrow aAb \mid \varepsilon$

$B \rightarrow b \mid bB$

$L = \{a^n b^{n-3} \mid n \geq 3\}$

| n | $a^n b^{n-3}$ |
|---|---|
| 3 | $a^3 b^0 = aaa$ |
| 4 | $a^4 b^1 = aaaab$ |
| 5 | $a^5 b^2 = aaaaabb$ |
| 6 | $a^6 b^3 = aaaaaabbb$ |

eq no. of as 3 as eq no. of bs

$S \rightarrow AB$

$A \rightarrow aaa \mid aAb \mid aAB$

$B \rightarrow \varepsilon \mid aBb$

~~$L = \{0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0\}$~~

1) $L = \{\underset{A}{0^m} \underset{B}{1^m 2^n} \mid m \geq 1 \text{ and } n \geq 0\}$

01, 0011 2, 000111 22 ...

$S \rightarrow AB$

$A \rightarrow 01 \mid 0A1$ n no. of 0 and n no. of 1s

$B \rightarrow \varepsilon \mid 2B$

2) $L = \{a^n b^m c^k \mid n + 2m = k \text{ for } n \geq 0 \text{ and } m \geq 0$

$a^n b^m c^k$

$a^n b^m c^{n+2m}$

$a^n b^m c^{2m} c^n$

$a^n \underset{A}{b^m c^{2m}} c^n$

means for every bs extra 2 cs are there

n no. of as and n no. of cs

$S \rightarrow aSc \mid A$

$A \rightarrow bAcc \mid \varepsilon$ } productions.

$V = \{S, A\}$

$T = \{a, b, c\}$

3) $L = \{w \mid |w| \bmod 3 \neq |w| \bmod 2 \text{ on } \Sigma = \{a\}$

$L = \{aa,$ 2 as

|w| length of string

$0 \bmod 3 \neq 0 \bmod 2$

$0 \neq 0$ false ✗

$1 \bmod 3 \neq 1 \bmod 2$

$1 \neq 1$ false ✗

$2 \bmod 3 \neq 2 \bmod 2$

$2 \neq 0$ ✓

$3 \bmod 3 \neq 3 \bmod 2$

$0 \neq 1$ ✓

$4 \bmod 3 \neq 4 \bmod 2$

$1 \neq 0$ ✓

$5 \bmod 3 \neq 5 \bmod 2$

$2 \neq 1$ ✓

$6 \bmod 3 \neq 6 \bmod 2$

$0 \neq 0$ ✗

$L = \{aa, aaa, aaaaa, aaaaa, aaaaa \ldots\}$

$S \rightarrow aa \mid aaa \mid aaaa \mid aaaaa \mid aaaaaas$

$6+2=8$

$6+3=9$

$6+4=10$

$6+5=11$

$L = \{w : |w| \bmod 3 \geq |w| \bmod 2\}$ on $\Sigma = \{a\}$

$0 \bmod 3 \geq 0 \bmod 2$

$0 \geq 0$ ✓

$1 \bmod 3 \geq 1 \bmod 2$

$1 \geq 1$ ✓

$2 \bmod 3 \geq 2 \bmod 2$

$2 \geq 0$ ✓

$3 \bmod 3 \geq 3 \bmod 2$

$0 \geq 1$ ✗

$4 \bmod 3 \geq 4 \bmod 2$

$1 \geq 0$ ✓

$5 \bmod 3 \geq 5 \bmod 2$

$2 \geq 1$ ✓

$6 \bmod 3 \geq 6 \bmod 2$

$0 \geq 0$ ✓

$7 \bmod 3 \geq 7 \bmod 2$

$1 \geq 1$ ✓

$S \rightarrow aaS \mid \varepsilon \mid a \mid aa \mid aaaa \mid aaaaa \mid aaaaaas$

## Derivation

Defination of derivation:

Let $A \rightarrow \alpha B \gamma$ and $B \rightarrow \beta$ are productions in grammer G where $\alpha, \beta, \gamma$ are string of terminals and/or nonterminals

A and B are non-terminals.

Then A produces the string $\alpha\beta\gamma$ by Replacing non terminals B by $\beta$ $\quad A \Rightarrow \alpha\beta\gamma$

This process of obtaining string of terminals and/or non-terminals from start symbol by applying some or all productions is called derivation.

$A \overset{*}{\Rightarrow} \alpha\beta\gamma$ 0 or more production

$A \overset{+}{\Rightarrow} \alpha\beta\gamma$ 1 or more production

$\Rightarrow$ derivation symbol

$E \to E+E$
$E \to E-E$
$E \to E*E$
$E \to E/E$
$E \to id$
identifier

$E \Rightarrow_{lmd} E+E$
$\Rightarrow id+E$
$\Rightarrow id+E*E$
$\Rightarrow id+id*E$
$\Rightarrow id+id*id$

lmd → left most derivation

sentential form

sentence

only terminals → sentence

terminal + nonterminals → sentential forms

## Left most derivation

In the derivation process of a left most variable is replaced at every step then derivation is said to be left most.

## Right most derivation

In the derivation process of a right most variable is replaced at every step then the derivation is said to be right most.

$E \Rightarrow_{rmd} E*E$
$\Rightarrow E*id$
$\Rightarrow E+E*id$
$\Rightarrow E+id*id$
$\Rightarrow id+id*id$

or

$E \Rightarrow_{rmd} E+E$
$E+E*E$
$E+E*id$
$E+id*id$
$id+id*id$

Obtain the left most derivation for the string aaabbabba for the following grammer $S \to aB/bA$, $A \to aS/bAA/a$, $B \to bS/aBB/b$

(✓ should start with S)

$S \Rightarrow_{lmd} aB$
$a\,aBB$ — as 1st letter ✗b ∴ ✗bA ∴ aB
$aa\,aBB\,B$
$aaab\,BB\,bS\,B$ — we can replace with either b or bS
$aaabb\,aB\,B$
$aaabba$ ; now we wont get so we have take B→b not B→bS

aaabbabbba

$S \xrightarrow{1\text{and}} aB$

$\Rightarrow aaBB$

$\Rightarrow aaaBBB$

$\Rightarrow aaabbB$

$\Rightarrow aaabbaBB$

$\Rightarrow aaabbabS$

$\Rightarrow aaabbabbA$

$\Rightarrow aaabbabbba$

*

Derivation tree (Parse tree)

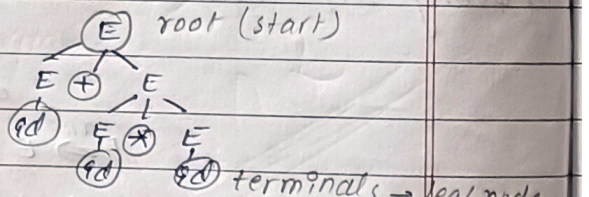
Let $G=(V,T,P,S)$ of a ~~cont~~ CFG it has following properties

1) The root has label of S

2) Every vertex has a label which is in variables, Terminals & epsilon

3) Every leaf node has a label from T and ~~Integer~~ interior vertex has a label from V

4) If a vertex is labled A of $x_1, x_2 \dots x_n$ are children of A Then $A \to x_1 x_2 \dots x_n$ in P




Partial Parse tree

(Partial derivation tree)

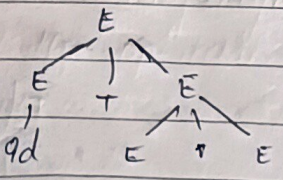
Let $G=(V,T,P,S)$ be a content free grammer the partial derivation tree has the following properties

1) The root has label S

2) Every vertex has a label which is in variables, Terminals & epsilon

3) Every leaf node has a label from variables, Terminals or epsilon (only this point is change

4) If a vertex is labelled A and of $x_1, x_2, \dots x_n$ are children of A then $A \to x_1, x_2, x_3 \dots x_n$ must in P.



# Ambiguous grammer

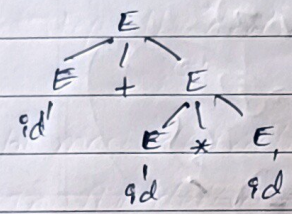
Let $G=(V,T,P,S)$ be a context free Grammer (CFG) A grammer G is ambiguous if and only if there exist only one string $w \in T^*$ for which 2 or more parse string exist by applying either left most derivation or right most derivation twice

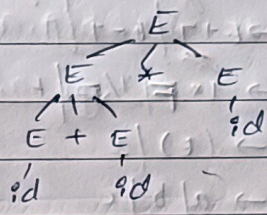
If we get 2 parse tree for string by applying LMD or RMD

1. $E \to E+E \mid E-E \mid E*E \mid E/E \mid id$ ambiguous? $id + id * id$

$E \xRightarrow{lmd} E+E$
$\Rightarrow id + E$
$\Rightarrow id + E*E$
$\Rightarrow id + id * E$
$\Rightarrow id + id * id$

$E \xRightarrow{lmd} E*E$
$\Rightarrow E+E*E$
$\Rightarrow id + E*E$
$\Rightarrow id + id * E$
$\Rightarrow id + id * id$



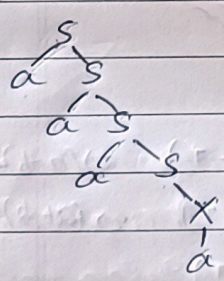

we got 2 diff parse tree
∴ ambiguous grammer

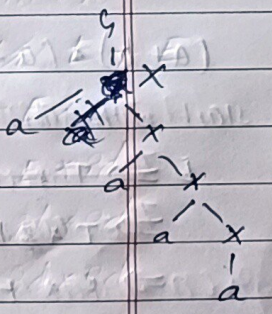
2. $S \to aS \mid X$ Check the following gramar is ambiguous or not
$X \to aX \mid a$
"aaaa"

take same

$S \xRightarrow{lmd} aS$
$\Rightarrow aaS$
$\Rightarrow aaaS$
$\Rightarrow aaaX$
$\Rightarrow aaaa$

$S \xRightarrow{lmd} X$
$\Rightarrow aX$
$\Rightarrow aaX$
$\Rightarrow aaaX$
$\Rightarrow aaaa$



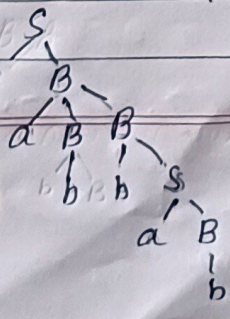




3. $S \to aB \mid bA$ "aabbbab"
$A \to aS \mid bAA \mid a$
$B \to bS \mid aBB \mid b$

$S \Rightarrow aB$
$\Rightarrow aaBB$
$\Rightarrow aabB$
$\Rightarrow aabbS$
$\Rightarrow aabbaB$
$\Rightarrow aabbab$

$S \Rightarrow aB$
$S \Rightarrow abS$
$S \Rightarrow abbA$
$S \Rightarrow abba$

$S \Rightarrow aB$
$\Rightarrow aaBB$
$\Rightarrow aabSB$
$\Rightarrow aabbAB$
$\Rightarrow aabbabB$
~~$\Rightarrow aabbaba$~~
$\Rightarrow aabbab$



aabbab → correct

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3. $S \to qCts / qCtses / a$ "qbtqbtaea"

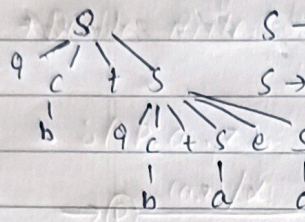
$C \to b$

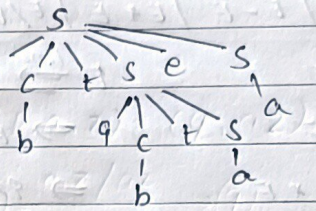
$S \to qCts$ | $S \to qCtses$

$S \to qbt * qCts$ | $S \to qbtS$ | $S \to qbtqCtses$

$S \to a$ | $S \to qbtqbtSes$ | $S \to qbtqbtaeS$

$S \to qbtqbtaea$





Obtain the unambiguous grammer for the grammer given

$E \to E+E / E-E$

$E \to E*E / E/E$

$E \to (E) / I$

$I \to a / b / c$

$E \to E+T / E-T / T$

$T \to T*F / T/F / F$

$F \to (E) / I$

$I \to a / b / c$

priority increases

Ambiguous grammer — Unambiguous grammer

$(a+b)*(a-b)$

start with start symbol (E)

$E \Rightarrow T$

$\Rightarrow T*F$ $\Rightarrow (I+T)*F$ $\Rightarrow (a+b)*(E-T)$ $\Rightarrow (a+b)*(a-I)$

$\Rightarrow F*F$ $\Rightarrow (a+T)*F$ $\Rightarrow (a+b)*(T-T)$ $\Rightarrow (a+b)*(a-b)$

$\Rightarrow (E)*F$ $\Rightarrow (a+F)*F$ $\Rightarrow (a+b)*(F-T)$

$\Rightarrow (E+T)*F$ $\Rightarrow (a+I)*F$ $\Rightarrow (a+b)*(I-T)$

$\Rightarrow (T+T)*F$ $\Rightarrow (a+b)*F$ $\Rightarrow (a+b)*(a-T)$

$\Rightarrow (F+T)*F$ $\Rightarrow (a+b)*E$ $\Rightarrow (a+b)*(a-F)$

*Assume + and - are having highest priority compared to * and / ~~then all the~~ and all the operators are left associative.

$E \to E*T / E/T / T$

$T \to T+E / T-F / F$

$F \to (E) / I$

$I \to a / b / c$

~~E → E*T | E/T | T~~

* Assume + and - are having highest priority compared to * & / and all operators are right associative

E → T+E | T/E | T

T → F+T | F-T | F

F → (E) | I

I → a | b | c

PUSH-DOWN AUTOMETA

Push-down Autometa is seven tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$

Q - set of finite states

Σ - set of

Γ - set of stack

δ - $Q \times (\Sigma \cup \varepsilon) \times \Gamma$ to finite subset of $Q \times \Gamma^*$

$q_0 \in Q$ start state

$z_0 \in \Gamma$ Initial symbol on the stack

$F \subseteq Q$ set of final states

PDA:

$\delta: Q \times \Sigma \cup \varepsilon \times \Gamma$ to $Q \times \Gamma^*$

The transition function of push down autometa depends on current state, the next input symbol on the top of the stack, the action performed by push down autometa consists of

*1 Changing the state from one state to another

*2 Replacing the symbol on the stack

δ(state, input-symbol, stack-symbol) = (next state, stack symbol)

$\delta(p, a, z) = (q, az)$ — here p moves from p → q by taking a

$\delta(p, a, z) = (q, \varepsilon)$ — here p → q by taking a by deleting z

$\delta(p, a, z) = (q, r)$ — z replaced by r

$\delta(p, \varepsilon, z) = (q, r)$ — here p → q by taking ε by replacing z with r

$\delta(p, \varepsilon, \varepsilon) = (q, z)$ — ↑ Initially stack empty

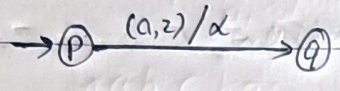
if p → q by ε and pushing z to stack

$\delta(p, \varepsilon, z) = (q, \varepsilon)$

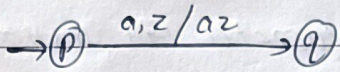
Graphical representation of PDA

* Nodes or circles → states of PDA
* Circle with an arrow i.e not originated from any other node → start state of PDA
* Double circles → final state of PDA

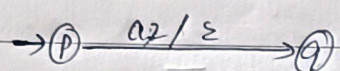
1. $\delta(p, a, z) = (q, \alpha)$



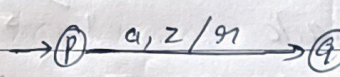

2. $\delta(p, a, z) = (q, az)$



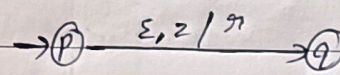

3. $\delta(p, a, z) = (q, \varepsilon)$




4. $\delta(p, a, z) = (q, \pi)$




5. $\delta(p, \varepsilon, z) = (q, \pi)$




Instantaneous Description

~~The current condition of Push Down~~

The current configuration of pushdown autometa at any given instant can be described by Instantaneous Description (ID)

An ID gives current state of PDA. The remaining string to be procesed and entire contents of the stack

An ID is given as 3 tuple - $(q, w, \alpha)$

$q \rightarrow$ current state

$w \rightarrow$ string

$\alpha \rightarrow$ contents of the stack

Left most symbol in the string $\alpha$ is on the top of the stack

Rightmost symbol in the string $\alpha$ is in the bottom of the stack

Current configuration of PDA is $(q, aw, z\alpha)$

$q \rightarrow$ c.s (current state)

$aw \rightarrow$ string to be procesed

$z\alpha \rightarrow$ current content of stack

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# ACCEPTANCE OF LANGUAGE BY PDA

* There are two cases where string w is accepted by Pushdown automata (PDA)

① Get the final state from the start state

② Get an empty stack from the start state

*** Let $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ be a PDA

language $L(M)$ accepted by a final state is defined as

$$L(M) = \{w \mid (q_0, w, z_0) \vdash^* (p, \varepsilon, \alpha)\}$$

$\alpha \in \Gamma^*$

$p \in F$

$w \in \Sigma^*$

language ~~$N(w)$~~ $N(M)$ accepted by an empty stack (null stack)

~~$N$~~ $N(M) = \{w \mid (q_0, w, z_0) \vdash^* (p, \varepsilon, \varepsilon)\}$

$w \in \Sigma^*$

$q_0, p \in Q$

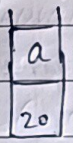
Obtain a PDA to accept the language $L(M) = \{wcw^R \mid w \in (a+b)^*\}$ where $w^R$ is reverse of w by a final state

$w = abb$, $w^R = bba$ $\Big\}$ $wcw^R$

$= abbcbba$, ~~or~~ $baacaab$ ~~or~~ - -

assume $z_0$ as final symbol

$\delta(q_0, a, z_0) = (q_0, az_0)$



on adding a

$q_0 \rightarrow q_0$

now stack has $az_0 \therefore (q_0, az_0)$

assume $z_0$ as final symbol

$\delta(q_0, b, z_0) = (q_0, bz_0)$

~~$\delta(q_0, a, a) =$~~

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, b) = (q_0, bb)$

stack

stack

$\delta(q_0, c, z_0) = (q_1, z_0)$

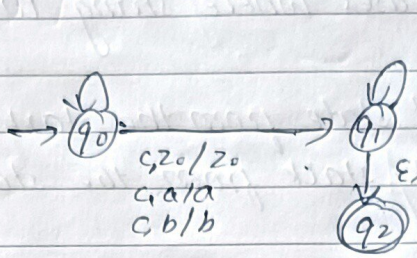
$\delta(q_0, c, a) = (q_1, a)$

$\delta(q_0, c, b) = (q_1, b)$

$\delta(q_1, a, a) = (q_1, \varepsilon)$

$\delta(q_1, b, b) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, z_0) = (q_1, z_0)$




b,b / bb
a,b / ab
a,z0 / az0
b,z0 / bz0
a,a / aa
b,a / ba

aabcbaa

$(q_0, aabcbaa, z_0) \vdash (q_0, abcbaa, az_0)$ (stack)

$\vdash (q_0, bcbaa, aaz_0)$

$\vdash (q_0, cbaa, baaz_0)$

c encountered change state $\vdash (q_1, baa, baaz_0)$

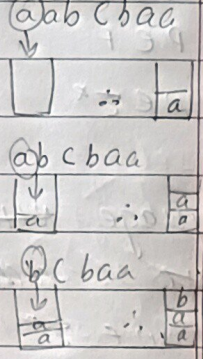
pop $\vdash (q_1, aa, aaz_0)$

$\vdash (q_1, a, az_0)$

$\vdash (q_1, \varepsilon, z_0)$

$\vdash (q_2, \varepsilon, z_0)$ now as reached final state

∴ string accepted




∴ $(q_0, aabcbab, z_0) \vdash (q_0, abcbab, az_0)$

$\vdash (q_0, bcbab, aaz_0)$

$\vdash (q_0, cbab, baaz_0)$

$\vdash (q_1, bab, baaz_0)$

$\vdash (q_1, ab, aaz_0)$

$\vdash (q_1, b, az_0)$ now not reached final state

string rejected

Obtain the pushdown automata to accept the language $L = \{a^n b^n \mid n \geq 1\}$ by final state

"aaabbb"

$\delta(q_0, a, z_0) = (q_0, az_0)$

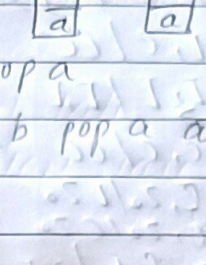
$\delta(q_0, a, a) = (q_0, aa)$

~~$\delta(q_0, a, a) = (q_0, aa)$~~

$\delta(q_0, b, a) = (q_1, \varepsilon)$

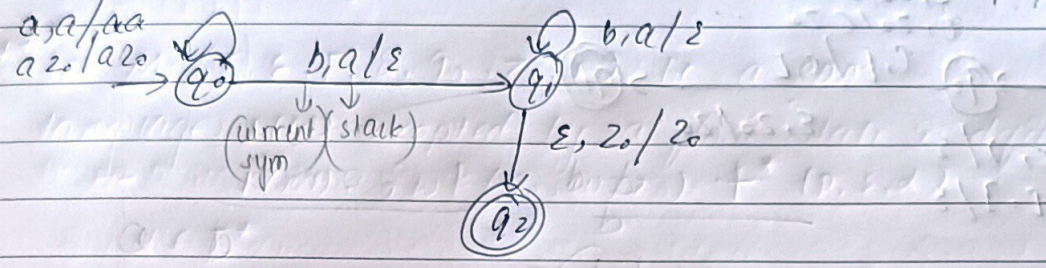
$\delta(q_1, b, a) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$




for every b pop a as no. of a = no. of b

pop all $\therefore \varepsilon$.




$(q_0, aaabbb, z_0) \vdash (q_0, aabbb, az_0)$

$\vdash (q_0, abbb, aaz_0)$

$\vdash (q_0, bbb, aaaz_0)$

$\vdash (q_1, bb, aaz_0)$

$\vdash (q_1, b, az_0)$

$\vdash (q_1, \varepsilon, z_0)$

$\vdash (q_2, \varepsilon, z_0)$

Obtain a pda to accept the language

$L(M) = \{w \mid w \in (a+b)^* \text{ and } n_a(w) = n_b(w)\}$

$\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

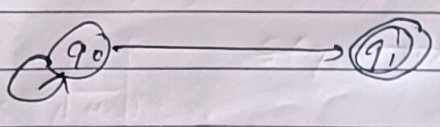
$\delta(q_0, b, a) = (q_0, \varepsilon_0)$ then pop as diff

$\delta(q_0, a, a) = (q_0, aa)$ then push as same

$\delta(q_0, a, b) = (q_0, \varepsilon_0)$ then pop as diff

$\delta(q_0, b, b) = (q_0, bb)$ then push as same

$\delta(q_0, \varepsilon, z_0) = (q_1, z_0)$




$(q_0, abbbaa) \vdash (q_0, bbbaa, az_0)$

$\vdash (q_0, bbaa, bz_0)$

$\vdash (q_0, baa, bbz_0)$

$\vdash (q_0, aa, bbbz_0)$

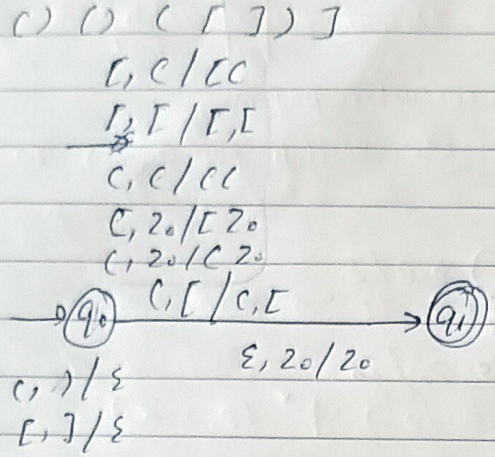
$\vdash (q_0, a, bbz_0)$

$\vdash (q_0, \varepsilon, z_0)$

$\vdash (q_1, \varepsilon, z_0)$

Obtain pda to accept the string of balanced paranthesis
to be considered on (,), [, ],

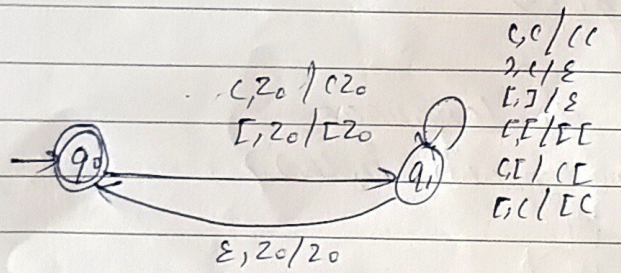
[ ( ) ( ) ( [ ] ) ]




$\delta(q_0, (, z_0) = (q_1, (, z_0)$
$\delta(q_0, [, z_0) = (q_1, [, z_0)$
$\delta(q_1, (, () = (q_1, (()$
$\delta(q_1, (, [) = (q_1, ([)$
$\delta(q_1, [, () = (q_1, [()$
$\delta(q_1, [, [) = (q_1, [[)$

$\delta(q_1, ), () = (q_1, \varepsilon)$
$\delta(q_1, ], [) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, z_0) = (q_0, z_0)$
as empty string shift state = final state

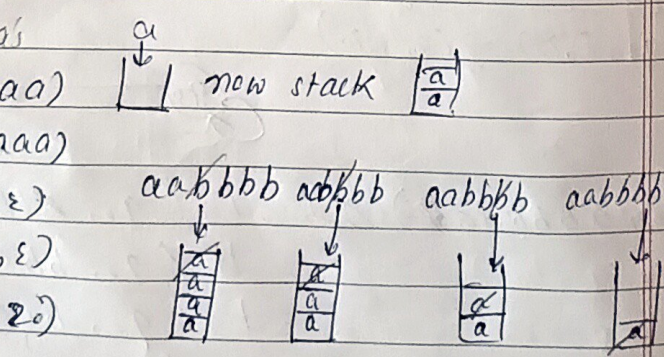
open paranthesis → push
close paranthesis → pop
} to keep record




$L = \{ a^n b^{2n} / n \geq 1 \}$
$L = \{ abb, aabbbb, aaabbbbbb, \ldots \}$

$a^n b^{2n}$ means for every a we add 2 extra as
to keep track, for each encountered a
we are adding 2 as

$\delta(q_0, a, z_0) = (q_0, aa)$
$\delta(q_0, a, a) = (q_0, aaa)$
$\delta(q_0, b, a) = (q_1, \varepsilon)$
$\delta(q_1, b, a) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$



# DETERMINISTIC & NON DETERMINISTIC PUSH-DOWN AUTOMETA

Let $M = (S, \Sigma, \Gamma, \delta, q_0, z_0, F)$ be a PDA

PDA is deterministic if

(1) $\delta(q, a, z)$ has only one element

(2) $\delta(q, \varepsilon, z)$ is not empty then $\delta(q, a, z)$ should be empty

Q1. Is the PDA to accept language $L(M) = \{ w c w^R \mid w \in (a+b)^* \}$ is deterministic?

$\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, a) = (q_0, bb)$

$\delta(q_0, c, z_0) = (q_1, z_0)$

$\delta(q_0, c, a) = (q_1, a)$

$\delta(q_0, c, b) = (q_1, b)$

$\delta(q_1, a, a) = (q_1, \varepsilon)$

$\delta(q_1, b, b) = (q_1, \varepsilon)$

~~$\delta(q_0, c, b) = (q_1, b)$~~

$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$